Multi-Label MeSH Term Classification of Biomedical Articles Using Pre-trained Language Models

Panteleimon Stanimeros
MSc in AI and Data Analytics, University of Macedonia
Thessaloniki, Greece
aid26006@uom.edu.gr

Nikolaos Strafiotis
MSc in AI and Data Analytics, University of Macedonia
Thessaloniki, Greece
aid26012@uom.edu.gr

Abstract

We address the task of automatic MeSH (Medical Subject Headings) term assignment to biomedical articles, formulated as a multi-label classification problem. Using only the title and abstract of each article, we fine-tune BioBERT, a domain-specific pre-trained language model, on the BioASQ 2015b Task A dataset. We evaluate performance using micro-averaged and macro-averaged F1 scores. Our results show that …

Keywords

biomedical text classification, multi-label classification, MeSH terms, BioBERT, BioASQ

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1 Introduction

Medical literature is indexed in PubMed using MeSH (Medical Subject Headings), a controlled vocabulary maintained by the National Library of Medicine. Manually assigning MeSH terms to the growing volume of biomedical publications is time-consuming and expensive. Automatic biomedical indexing, the task of predicting MeSH terms from an article's text, therefore has significant practical value.

This paper presents our approach to the BioASQ Task A challenge [9], which frames the problem as multi-label classification: each article may be assigned multiple MeSH terms simultaneously. We focus exclusively on the title and abstract as input, and leverage pre-trained biomedical language models, specifically BioBERT [5], to learn contextual representations.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our approach. Section 4 presents experimental results. Section 5 concludes.

2 Related Work

Early approaches to biomedical article indexing relied on TF-IDF weighted bag-of-words representations combined with k-nearest neighbours (MTI, [7]) or SVMs. The introduction of word embeddings, such as Word2Vec [6] and fastText [4], improved performance by capturing semantic similarity between medical terms.

More recently, transformer-based models pre-trained on large biomedical corpora have shown strong results. BioBERT [5] and SciBERT [1] both extend BERT [2] with domain-specific pre-training and have achieved state-of-the-art performance on multiple biomedical NLP benchmarks. In the BioASQ challenge, fine-tuned transformer models consistently outperform shallow approaches, with MeSH-CNN [3] and AttentionMeSH [8] being notable prior systems.

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3 Method

3.1 Problem Formulation

Given the title t and abstract a of a biomedical article, the goal is to predict a subset $\hat{Y} \subseteq L$ of MeSH labels from the full label set L. This is a multi-label classification problem with $|L|$ binary outputs.

3.2 Input Representation

We concatenate title and abstract using the `[SEP]` token:

$$x = \texttt{[CLS]}\ t\ \texttt{[SEP]}\ a\ \texttt{[SEP]}$$

The resulting sequence is truncated to 512 tokens, the maximum context length of BERT-based models.

3.3 Model Architecture

We fine-tune BioBERT (`dmis-lab/biobert-base-cased-v1.2`) as our encoder. The `[CLS]` token representation from the final layer is passed through a dropout layer and a linear classification head:

$$\hat{y} = \sigma(W \cdot \text{Dropout}(h_{\texttt{CLS}}))$$

where $W \in \mathbf{R}^{|L| \times d}$, $d = 768$ is the hidden size, and σ is the sigmoid function applied element-wise. We use binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N|L|} \sum_{i=1}^{N} \sum_{j=1}^{|L|} \left[y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log(1 - \hat{y}_{ij}) \right]$$

3.4 Training Details

We use the AdamW optimiser with a learning rate of 2×10^{-5} and gradient clipping at 1.0. MeSH labels appearing fewer than 10 times in the training set are discarded to reduce the output vocabulary and mitigate extreme class imbalance. Labels are predicted at inference time by applying a threshold $\tau = 0.5$ to the sigmoid outputs.

4 Experiments

4.1 Dataset

We use the BioASQ 2015b Task A training dataset (v.2015b), which contains 4,607,922 PubMed articles annotated with MeSH 2015 terms. Table 1 summarises dataset statistics.

Table 1: BioASQ 2015b dataset statistics.

Statistic	Value
Total articles	4,607,922
Avg. MeSH labels per article	13.08
Unique MeSH terms	26,866
Train / Val split	90% / 10%

4.2 Evaluation Metrics

Following BioASQ evaluation practice, we report micro-averaged F1 (favours frequent labels) and macro-averaged F1 (treats all labels equally).

4.3 Results

Table 2: Results on the BioASQ 2015b validation set.

Model	Micro-F1	Macro-F1
TF-IDF + kNN (baseline)	–	–
BioBERT (ours)	–	–

5 Conclusion

We presented a fine-tuned BioBERT approach for multi-label MeSH term classification on the BioASQ Task A dataset. …

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